

Development of Woven Fabric-based Electrical Circuits

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ABSTRACT

This paper describes the development of woven electrical circuits, which are formed by interlacing conducting and non-conducting threads into a woven fabric. Conductive threads in these electrical networks are arranged and woven such that they follow desired electrical circuit designs. Electronic devices can be attached to these electrical networks, which can serve as flexible circuit boards. In these woven circuits, an efficient transfer of current from one conductive yarn to an orthogonal one is achieved by the formation of an effective electrical interconnect at the point of intersection of these yarns. Formation of woven conductive networks also involves disconnect formation or cutting of conductive yarns at certain specified points. Different methods and processes were identified and applied in order to form interconnects and disconnects at specified points of these fabrics. Efficacy of these interconnects was evaluated by DC resistance and AC Signal measurements. The results of these evaluations are reported. The conductive threads woven into these fabric-based circuits were also evaluated for signal integrity issues.

INTRODUCTION

Woven fabric structures offer numerous opportunities to be used as a platform to incorporate electronic devices due to their structural order, low flexural rigidity and their ability to conform to almost any desired shape. Using today's technologies, intricate woven fabric structures can be manufactured continuously at high speed with low production costs. Some work has been done previously on the integration of conductive elements and electronic devices into textiles [1-6]. These textile-based electrical circuits and devices could be used in the fields of medicine, communications, electronics, automobiles, space exploration, and several military applications.

The creation of woven electrical circuits or circuit boards not only requires the formation of interconnects at different crossover points but also disconnects at predetermined locations along a conducting yarn [1] so as to control the current path in the circuitry. This is shown schematically in Figure 1 where several chips need to be connected via a fabric. In this paper we focus on the formation of interconnects and disconnects made by resistive welding after conductors have been woven into a fabric.

Signal integrity issues in fabric-based woven circuits are also important. When conductive thread elements (which can be modeled as transmission lines) are placed parallel to each other in these woven circuits, it can lead to capacitive and inductive signal crosstalk between the neighboring lines and examples of this are also presented.

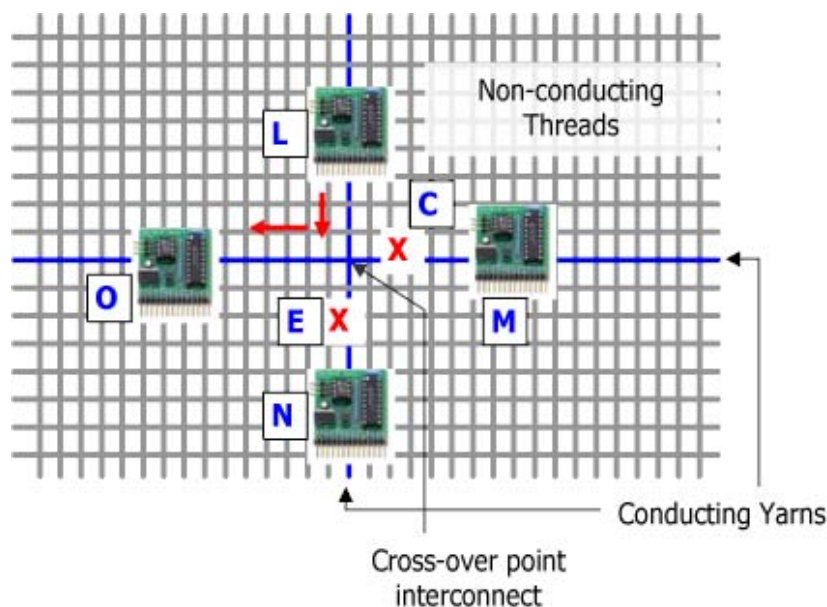


Figure 1. Schematic of a woven electrical network having crossover point interconnects and disconnects. Red line represents multiple conductive yarns connected to chips via micro welding.

CROSSOVER-POINT INTERCONNECTS AND DISCONNECTS

The main objective of crossover-point interconnect formation is to reduce the DC resistance associated with the interconnect and is achieved by selecting the correct material specific bonding/welding procedure. Different interconnect techniques have been identified as potential methods to form interconnect at crossover points of woven based circuits. These are resistance welding, air splicing, ultrasonic welding/bonding, laser beam welding, conductive adhesive bonding, microwave bonding, and solvent bonding. In this paper, evaluation of resistance welding as a method of formation of crossover point interconnects in woven fabric-based electrical circuits is discussed.

Resistance welding is a process in which heat is generated at the interface of the components to be joined by passing an electrical current through the components. The electric current is applied for a fixed length of time while an appropriate level of force is applied at the interface of the components to be welded to form the bond. The heat generated in resistance welding is a function of I , the weld current through the conducting threads to be welded, R , the electrical resistance (in ohms) of the conducting threads, M , a thermal constant and a function of conductive thread geometry and weld force, and t , the time (in milliseconds or microseconds) of current passage. The resistance R includes the bulk resistances of the electrodes and the threads to be fused, the contact resistances between the electrodes and the conductive threads and the contact resistance between the conductive threads themselves. These resistances are in series and can be summed up to give the net resistance R encountered during welding. In this study Top-Bottom welding methods are compared with Parallel gap methods, the two methods being defined by the electrode placement with respect to the fabric.

Fabrication of fabric-based circuit boards also involves formation of disconnects to control the current path in the circuitry. As shown in Figure 1, if the current flowing from point O has to be prevented from reaching the devices placed at points M and N, then disconnects have to be produced at points C and E respectively. Disconnects can be formed by creating a fabric with

floats at certain points. A float can be defined as a thread in a woven fabric, which goes over two or more orthogonal threads without being interlaced. These floats could be selectively cut by cutters (or micro-cutters) to form the disconnects. Disconnects in woven fabric-based circuits can also be produced by parallel gap resistance welding. When large values of weld currents are used in this welding process, a large amount of heat is generated, which melts the thread material between the probes, thereby creating cuts or disconnects.

EXPERIMENTAL DETAILS

The conductive threads used for the formation of woven circuits were bare steel yarns (2-ply steel yarns; each yarn having 275 fibers, each of 12 denier (i.e. mass of 9000 meters of each fiber is 12 grams) and bare copper yarns (40 strands twisted to give an overall diameter of 0.33 mm). These multi-filament yarns were very flexible and can be woven into a fabric like textile yarns. The insulating material used for the formation of these circuits were polyester yarns (polyester yarns of 440 denier (warp), polyester yarns of 140 denier (weft)). Plain-weave fabrics were woven from conducting twisted copper and steel yarns using a Eltex weaving machine equipped with a jacquard. The copper yarns were woven in warp and filling directions at certain intervals. The conductive yarns woven in the warp direction were mounted on a creel to maintain constant tension on the warp threads during weaving. In order to develop samples of crossover point interconnects for evaluation, fabric samples with one crossover point were cut out of the main fabric. Distance between end points of the sample was maintained at 5 inches. The crossover points of the cut out samples were welded using Top-bottom probe and Parallel gap methods.

In order to study AC signal Crosstalk between neighboring copper yarns, these yarns were woven into a fabric to get parallel interconnection lines. The conductive yarns used for the purpose of weaving parallel transmission lines were bare copper yarns (28 gauge). The weaving was done on a Eltex weaving machine having Staubli Jacquard head. The spacing (d) between two parallel sets of these transmission lines was varied with four sets of samples being produced – with 1/8th inch, 1/4th inch, 1/2 inch, and one inch spacings. 10-inch long samples were cut out of the fabric samples having woven copper yarns and these samples were evaluated for crosstalk noise. The oscilloscope used for crosstalk evaluation was a two-channel oscilloscope, Tektronix TDS-340. AC signals varying between 1 KHz and 10 MHz were sent on one of the two parallel transmission lines and the effect of the signal changes in the aggressor line, (when the signal goes from low to high or from high to low) on the neighboring quiet line (carrying no signal and spaced d inches apart) were evaluated by looking at the output waveforms of both the aggressor and the victim lines.

DISCUSSION

The resistance values (R) of different samples of welded copper fabrics are presented in Table 1. The resistance of a woven copper yarn sample without any weld was found to be significantly higher than that of a copper thread (not woven into the fabric). This is due to the presence of a crossover point. In the case of woven copper yarn samples (with and without the welds), when current flows through a crossover point, not all current (charge per unit time) may

Conductive Yarn/Fabric Samples	Resistance Welding Method	Weld Current, (Amps)	Normalized Resistance*	Standard Deviation
Unwelded Copper Fabric	-	-	9.4832	0.0317
Copper Yarn	-	-	1.0000	0.0009
Welded Copper Fabric	Top-Bottom Probe	1000	2.0641	0.0016
Welded Copper Fabric	Top-Bottom Probe	1400	1.2583	0.0017
Welded Copper Fabric	Top-Bottom Probe	1800	1.0000	0.0011
Welded Copper Fabric	Top-Bottom Probe	2100	1.3222	0.0010
Welded Copper Fabric	Parallel Probe	1000	1.3254	0.0275

Table I. Crossover point DC Resistance values for woven Copper fabric samples

* Actual Resistance Normalized with respect to the actual resistance of a continuous copper yarn (0.031 Ohms/5 inches)

be transferred from the current carrying conducting yarn to the orthogonal conducting yarn intersecting it at the crossover point. This leads to a higher value of measured resistance for these samples as compared to that for copper thread only. The DC resistance value for an isolated copper thread serves as a base-line curve for all the resistance evaluations.

When resistance welds (using top-bottom probes and parallel gap welding) were formed at the cross-over points, DC resistances of the welded samples were found to be less than those for woven copper yarn samples without any weld sample by 6-9 times. This could be attributed to a higher value of resistance at the crossover point contact between the two intersecting conducting yarns, in case of the woven copper yarn sample without any weld.

The value of DC resistance for top-bottom probe welded sample (welded with 1800 Amperes of weld current) is found to be less than that in case of top-bottom probe welded samples (welded with 1000 Amperes and 1400 Amperes of weld current). The greater amount of fusion or bonding created due to increase in weld current can be explained on the basis that heat generated during resistance welding (so as to cause the melting or fusion) is equal to I^2Rt , i.e., it is proportional to the square of the weld current. 1800 Amperes is the optimum weld current as the lowest value of crossover point interconnect resistance is achieved for this current and further increase in weld current does not lead to any reduction in this resistance.

The standard deviation in the DC resistance values (between different readings) for the copper thread in the free state and for the welded samples was found to be much lower than that for a woven copper yarn sample without any weld. This is one of the main reasons for creating a weld or a bond at the crossover point of the intersecting orthogonal yarns, as the weld or bond helps to overcome the variation in the current flow (as would be the case in an unwelded crossover point) when the conducting fabric is used in everyday life and is subject to bending and stretching.

The values of DC resistances are higher in the case of the parallel gap welded samples as compared to those for samples welded by top-bottom probe welding. The standard deviation in the resistance values of parallel gap welded samples was also found to be higher than that for the top-bottom probe welded samples. This shows that the top-bottom probe method is a preferred method for achieving a more efficient interconnect (lower and more uniform resistance associated with the crossover point interconnect) at the crossover point.

The welded and the unwelded copper fabric samples were analyzed under the SEM microscope to study the nature of interconnects at the crossover point. It can be seen from the pictures of parallel gap welded copper fabric samples (figure 2) that the individual conducting fibers fuse and form a bond at the crossover point wherever the passage of current takes place

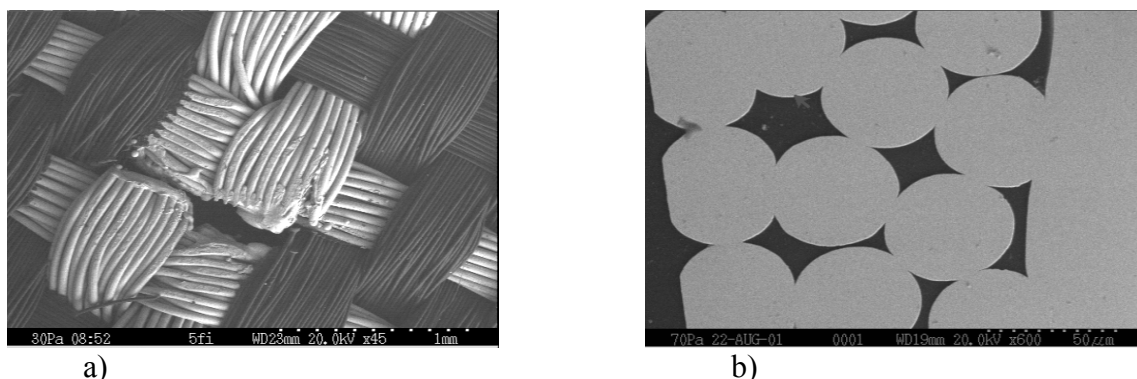


Figure 2. SEM micrographs of copper, (light) and polyester (dark) threads. In a) A disconnect has been formed with resistive welding. In b) SEM cross-section shows how individual fibers are welded together and to the orthogonal thread.

from one of the parallel gaps into the fibers/yarns and then from the fibers/yarns into the second arm of the parallel gap. In this process, the generation of heat to create melting/fusion of the copper conducting yarns at the crossover point can be controlled precisely to create weld formation at a specific point. This method, as seen in figure 2, cannot only be used to produce interconnects but also produce disconnects (discussed earlier) at the point where the parallel gaps contact the conducting fabric.

SEM pictures of welded samples showed that the damage to neighboring non-conducting fibers in top-bottom probe welding was greater for higher values of weld current and this can be attributed to greater amount of heat being generated on using a higher value of current. It was also observed that the extent of the damage to these fibers is much less in case of Parallel Gap Welding than that in case of Top-bottom probe welding.

The average value of DC resistance for a fabric sample with steel yarns and an unwelded crossover point was found to be almost 1.5-2 times the resistance in case of parallel gap welded samples. The standard deviation between the different readings for the welded steel samples were much lower than that for the unwelded woven steel sample. A higher value of resistance is obtained in the case of fabric samples with steel yarns/fibers as compared to those with copper yarns/fibers and this can be attributed not only to the higher bulk resistance of steel but also to a more efficient weld/interconnect being formed at the crossover point in the case of copper as compared to steel.

In evaluation of signal integrity in woven circuits, when a one Volt AC square wave (with a frequency of 1 MHz) was sent down a signal transmission line (woven bare copper yarn), one should not observe any voltage signal on the neighboring line (parallel to the signal line and spaced 1/8 inch apart) as it is a quiet line (assuming that there is no crosstalk). But one does observe (as shown in figure 3) a positive spike peak of 106 mV on the quiet line every time the voltage signal on the signal line goes from Low (0V) to High (1V) corresponding to the rising edge of the signal on the aggressor line. One also observes a negative spike trough of -106 mV on the quiet line every time the voltage signal on the signal line goes from High (1V) to Low (0V) corresponding to the falling edge of the signal on the aggressor line. Similar results were obtained for input waveform frequencies higher than 1 MHz and for other line spacings. This appearance of signal on a quiet line can be attributed to capacitive and inductive coupling between the signal and the quiet lines leading to existence of crosstalk between the lines. Crosstalk was observed to be higher for lower line spacings.

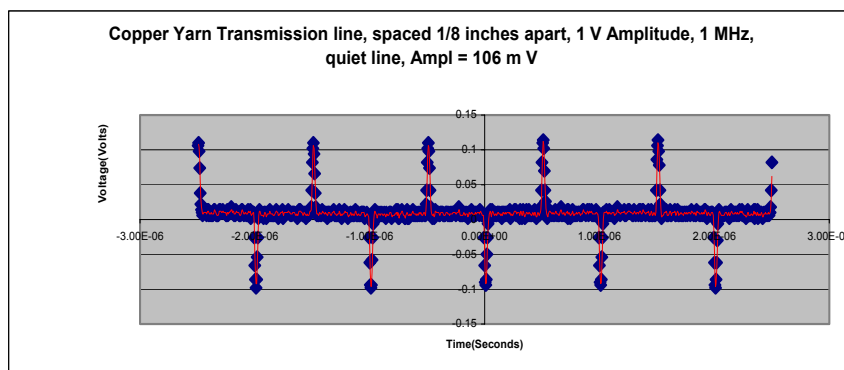


Figure 3. Voltage vs. time output waveform obtained on a neighboring quiet line (Bare copper yarn transmission line)

CONCLUSIONS

In this paper, woven fabric-based electrically conductive circuits have been described which consist of conductive threads woven into a fabric network along with the non-conducting threads. Weaving the conductive threads so as to follow a desired electrical circuit design was done and interconnects were developed at the crossover points of orthogonal conductive yarns. Resistance welding was identified as an effective methods of producing crossover point interconnects and was applied to produce these interconnects. The welded and unwelded samples were evaluated for their DC resistance values and it was observed that the resistance-welded crossover point interconnects had lower and more uniform resistance associated with the crossover point as compared to the unwelded samples.

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